Mario Markets cheat sheet

How it's won: the money is in the **1000-lot score contracts** (#1, #2, Top-2). At full size, 1 point of mispricing = 1,000 of PnL; the same point is worth 200 on a full series-winner position and 100 on a race winner. The biggest unknown is how good the 4 humans are against the 8 CPUs: it moves fair Top-2 anywhere from about 66 to 97. **Race 1 is the first real evidence: reprice the Series-2 score contracts in the first minute after it.** Then price #1 and Top-2 *above* the naive average (the order-statistic premium), price series-winner contracts from the lead tables, take every arbitrage, and **never leave quotes resting while a race runs.**

Rules that matter

Item	Rule
Points per race (overall place of 12)	1st 15, 2nd 12, then **13 – place** (3rd 10 ... 12th 1)
Race winner (limit 100)	Best-placed **human** = 100, even if a CPU wins the race. The 4 contracts always sum to exactly 100
Series winner (limit 200)	Highest 4-race total = 100. **Tied leaders each settle 50**, so a 3-way tie pays 150 in total
#1 / #2 / Top-2 (limit 1000)	Settle to the best human's series points, the 2nd best's, and their sum. 45 and 45 give 45 / 45 / 90
Characters	Daisy, Peach, Birdo and Yoshi have identical stats and the same kart. A favourite-character premium is noise

Pocket card (memorise)

Item	Value
Prior fair values	Before Race 1: race and series winner 25. **#1 ≈ 45, #2 ≈ 37, Top-2 ≈ 82** (sd 13.5). Don't bet the level yet
Robust prior range	#1 36-52 · #2 29-45 · **Top-2 65-97** · #1 – #2 spread 5-10. A price inside the range is a bet, not an edge
Selection premium, series start	**#1 = 4m + 8, Top-2 = 8m + 10**, where m = expected points per human per race (m ≈ 6.8 if humans are no better than the CPUs, 8.5 if the average human finishes 5th, 10.5 if strong). Uneven skill adds about +2 / +3
Human level	$\hat{m} = 8.5 + \beta(\bar{x} - 8.5)$, $\bar{x}$ = average human points per race so far. **β = 0.45, 0.6, 0.7, 0.75** after races 1-4 (then 0.8-0.85)
One human's skill	$m_i = \hat{m} + \gamma(\bar{x}_i - \bar{x})$. **γ = 0.25, 0.4, 0.5, 0.55** after races 1-4 (then 0.6-0.7)
Series-2 Top-2	During Series 1: **82 + k($\bar{x}$ – 8.5), k = 3.1, 4.4, 5.1, 5.5** after races 1-4
Two-human duel	Leader ≈ 100·Φ(lead / (5.5·√races left))
Locks	Only with one race left: a **lead of 15+ locks** the series (14 can only tie)
Finish line	Pre-loaded tickets **buy ≤ 90, sell ≥ 10**. Never market orders

Right after Race 1: the key repricing (T = the four humans' total Race-1 points)

Race 1 total T	Read	Series-2 action (Series-1 #1 / Top-2 move about 0.75× as much)
≥ 42	Strong humans	S2 Top-2 fair 88-91, #1 48-49. **Buy Top-2 offered ≤ 85, #1 ≤ 45.5**
35-41	Moderately strong	Top-2 fair 83-87. Trade only gaps ≥ 4
28-34	Little information	Stay near 78-82. Check whether one human dominates
22-27	Weak-ish	Top-2 fair 73-77. **Sell Top-2 bid ≥ 82**
≤ 21	Weak, or 1 strong + 3 weak	Top-2 fair 56-73. **Sell Top-2 bid ≥ 78, #2 bid ≥ 36**

- **Size:** immediately 30-40% of the limit on S2 Top-2, and 20-30% each on S2 #1 and #2. Add up to 60% after Race 2 if it confirms; cut back if it doesn't.
- **One dominant human** (e.g. places 1, 10, 11, 12): buy that human's S2 series contract up to fair – 5 (fair ≈ 59), ≤ 50-100 lots until Race 2 confirms. After a 1, 4, 7, 10 race their fair is ≈ 42.

Mid-series score fair values

Notation: r = races left; s1 ≥ s2 ≥ s3 ≥ s4 = current scores; a, b, c = gaps of 2nd, 3rd and 4th behind the leader; m_1, m_2 = the top two humans' own expected points per race.

- **$E[\#1] = s1 + r \cdot m_1 + C(a) + 0.45 \cdot [C(b) + C(c)]$**
- **$E[\text{Top-2}] = s1 + s2 + r \cdot (m_1 + m_2) + [C(b - a) + 0.5 \cdot C(c - a)] \times (1 + 0.55 \cdot C(a)/C(0))$**
- **$E[\#2] = E[\text{Top-2}] - E[\#1]$**

Selection premium versus "score + races left × m" (S2, equal skill):

	Series start (r = 4)	All tied, r = 3	All tied, r = 2	All tied, r = 1	Leader 13+ clear, r = 1
#1	+8.2	+7.1	+5.9	+4.5	≈ 0
Top-2	+10.4	+9.0	+7.4	+5.5	–

C(g), the expected amount a rival g points behind finishes ahead of you. Pick the level from m̂ (S1 = humans no better than CPUs, m ≈ 6.8; S2 = average, 8.5; S3 = strong, 10.5) and interpolate between levels:

r	Level	g = 0	1	2	3	4	5	6	8	10	12	15
1	S1	2.48	1.98	1.55	1.19	0.89	0.65	0.46	0.21	0.08	0.02	0
1	**S2**	**2.33**	**1.83**	**1.40**	**1.03**	**0.75**	**0.52**	**0.35**	**0.14**	**0.04**	**0.01**	**0**
1	S3	1.91	1.41	0.98	0.64	0.42	0.23	0.12	0.02	0.00	0.00	0
2	S1	3.38	2.91	2.48	2.09	1.75	1.45	1.18	0.76	0.47	0.27	0.10
2	**S2**	**3.14**	**2.67**	**2.25**	**1.87**	**1.54**	**1.25**	**1.00**	**0.61**	**0.35**	**0.19**	**0.06**
2	S3	2.52	2.05	1.65	1.30	1.00	0.76	0.56	0.27	0.11	0.04	0.00
3	S1	4.13	3.65	3.21	2.80	2.43	2.10	1.80	1.29	0.90	0.61	0.32
3	**S2**	**3.84**	**3.36**	**2.92**	**2.52**	**2.16**	**1.84**	**1.55**	**1.08**	**0.72**	**0.46**	**0.22**
3	S3	3.08	2.60	2.18	1.80	1.47	1.19	0.94	0.57	0.32	0.17	0.05

Series-winner value: leader against one rival (average humans, other two out of contention)

Lead	0	1	2	3	4	5	6	7	8	9	10	12	14	16
1 race left	50	53	60	68	74	80	85	89	93	95	97	99	99.9	100
2 races left	50	55	60	64	69	73	77	81	84	87	90	94	96	98
3 races left	50	54	58	62	66	70	73	76	79	82	85	89	93	95

- **The rival** is worth 100 minus the leader. **Strong humans:** leads are stickier; with 1 race left, a lead of 4 → 80, 6 → 91, 8 → 98.
- **A leader who has also looked stronger:** +3 to +6.
- **Chasers bunched together:** don't multiply duel values; use the full matrices in the playbook (§3.7 B).

Hard bounds and arbitrage (risk-free when quotes break them)

- **Bounds:** #1 is 10-60, and **56, 58 and 59 are impossible**. #2 is 8-54. Top-2 is 20-108, and **107 is impossible**.
- **Mid-series:** #1 ∈ [s1 + r, s1 + 15r]; Top-2 ∈ [s1 + s2 + 3r, s1 + s2 + 27r].
- **Out of reach:** one race moves two humans apart by at most 14, so a human trailing by more than 14r is worth 0.

Arbitrage. On multi-leg trades, do the stale leg first, sized to what you'd hold if the other legs miss.

Relationship	Violation → trade
Race basket = 100	Sum of bids > 100: sell all 4. Sum of asks < 100: buy all 4
Series basket ≥ 100	Sum of asks < 100: buy all 4. Sell only at ≥ 101 (≥ 102.5 if humans are strong), ≤ 50 baskets: a 3-way tie costs 49 per set
Top-2 = #1 + #2	Top-2 bid > #1 ask + #2 ask: sell Top-2, buy both. And the reverse
#1 ≥ #2	#2 bid > #1 ask: buy #1, sell #2
Same human, future races	Equal fair value: buy the cheapest, sell the richest
Exit trick	A long race contract plus the other three contracts of that race is a riskless 100. When the bid is thin, buy the other three instead of selling yours

Phase plan

1. Before Race 1

- **Scan for arbitrage.** Probe the house quotes: #1 quoted near 36 (the naive 4 × 9) means a big structural edge all day.
- **Quote wide and small,** both series the same:

	Race winner	Series winner	#1	#2	Top-2
Quote	21 / 29	20 / 30	40.5 / 49.5	33 / 41	76 / 88
Size (lots)	10-25	10-25	25-100	25-100	25-100

- **#1 – #2 spread:** buy at ≤ 4.5, sell only at ≥ 10.5, ≤ 300 lots.
- **Fan premia:** sell any character's race contract bid at ≥ 30.
- **60 s before the green light: cancel every quote** (nearly all depend on Race 1). Only stink quotes ≥ 8 points away on future contracts may stay.

2. During a race

- The Spotter calls places every ~5 s using quadrant codes ("Y2 D5 P7 B11").
- If score contracts trade live, Gun B reprices #1 and Top-2 at the start of the final lap. Trade edges ≥ 3, ≤ 200-300 lots in Series 1.
- **Finish line:** buy ≤ 90 / sell ≥ 10. Go to 95/5 only with two-person confirmation and no busts.
- When the last human crosses, score contracts = realised points + formula. Hit quotes ≥ 2 off.

3. Between races (aim for 60 s)

1. Record all four places.
2. Update $\hat{m}$ and m_i.
3. Reprice current-series scores (formulas above) and hit stale quotes ≥ 2 off at 30-50% of the limit.
4. Reprice Series-2 contracts.
5. Series winner from the lead table: edges ≥ 5 at 50-100 lots.
6. Scan for arbitrage.
7. Requote, then cancel before the green light.

Half-spread	Race winner	Series winner	#1 / #2	Top-2	Series 2
After Race 1	±3	±4	±2	±3	±4
After Race 3	±3	±2-3	±1	±1.5	±3

- **Fade the hot human:** if the last winner's next race is bid 45-50 when fair is ≈ 36, sell 25-50 lots.

4. Races 3-4 into Series 2

- Pre-compute the Series-2 grid: S2 Top-2 = 82 + 5.5·($\bar{x}$_S1 − 8.5), plus each human's skill.
- **Fire it the moment Series 1 ends.** Most teams look at Series 2 only after Series 1 settles.

In-race (only if markets trade during races; low confidence, so calibrate on Series 1)

Leading human's chance to win the race-winner market at the start of the final lap. p is their overall place, q the next human's place; the second number assumes a chaotic last lap. Gun A buys only when the offer is ≥ 10 below the table, 25-50 lots in Series 1.

p \ q	2	3	4	5	6-7	8-9
1	64/56	73/64	80/70	85/75	90/80	94/86
2		54/48	64/55	70/60	77/66	87/74
3			50/44	58/50	68/56	78/64
4				48/43	58/48	72/61

Expected final points from place at the start of the final lap:

Place	1	2	3	4	5	6	7	8	9	10	11	12
Points	13.3	11.5	10.2	9.3	8.4	7.7	6.9	6.2	5.4	4.6	3.7	2.6

A clear series leader sitting 1st is worth about +4.8 on #1 (13.3 against a pre-race 8.5). Otherwise scale the shift by the chance that human ends as #1 (or top two, for Top-2).

Risk, roles, tool

- **Orders:** limit orders only, priced to fill with a cap. Type 100/200/500 (the presets stop at 50). At most ~6 resting orders per trader, logged on paper.
- **Level cap:** L = S2 Top-2 + 0.5·(S2 #1 + S2 #2), plus (races left / 4) × the same for Series 1. Keep **|L| ≤ 300 before Race 1, ≤ 700 after Race 1, ≤ 1,000 after Race 2.**
- **One human's skill:** future race-winner lots + 2 × series-winner lots ≤ 400 net.
- **Reserve:** keep ≥ 300 of 1000 score capacity free for the Race-1 repricing.
- **Don'ts:** bet the level before Race 1 inside the range; pay 97-99 at the finish line; sell the series basket below 101; treat one hot race as proof of skill.
- **Roles:**
  - **Spotter:** watches only the race and calls places; never touches a keyboard. Writes the quadrant → character → contract mapping before every race.
  - **Gun A:** race winners, the finish line, arbitrage, between-race quotes.
  - **Gun B:** current-series score and series-winner contracts.
  - **Brain:** standings, $\hat{m}$/m_i, tables, Series-2 prices, caps, the log.
- **Ask on Slack:** Can the current race's contracts trade while it runs? Are position limits per contract? Can we short? Who provides liquidity? Same 4 racers in both series? Are trades busted after the outcome? Does Mario count toward the overall rank?

Tool (Brain's laptop, not the trading one). Run it from the `mario/` folder, with places and scores in contract order H1..H4 (not sorted); add `--K 200000` once there are 6+ races of history. It has no race-to-race shocks, so it overreacts: after Race 1, move about 85% of the way from the prior to its number (90% after Race 2).

```
python3 -I tools/fv_calc.py --hist "4,1,7,9;6,2,5,10" --scores 16 27 14 7 --left 2
```